

# Portfolio Fixes Production Now

Target: niranjanvsk.sxyz. Implementer: Opus/Fable. Keep every change surgical — touch only what's named below, no unrelated refactors, no design-system or copy drift outside scope.

## Resolved decisions (do not re-litigate, just implement)

- **No external CDN.** All loader/mindmap assets ship as local bundled files ( `public/` or static import). Vercel's edge network already CDNs static assets on deploy.
- **Mind map preloads in background.** During the initial loader's idle time, prefetch/hydrate the mind map's heavy deps ( `@xyflow/react` , `react-force-graph-3d` ) via a background `import()` so `/map` is warm on first navigation.
- **Chatbot model routing:** Vercel AI Gateway as the routing layer, **Gemini 3** as default model, **Groq** kept as fallback in the gateway config. No pipeline/architecture rewrite — swap only the provider the gateway calls.
- **Guardrails:** implement as versioned Markdown system-prompt files (no RAG pipeline). Strip any "using Groq key" / "will be integrated when actual key given" placeholder text from visible chatbot responses — that must never render to the user.

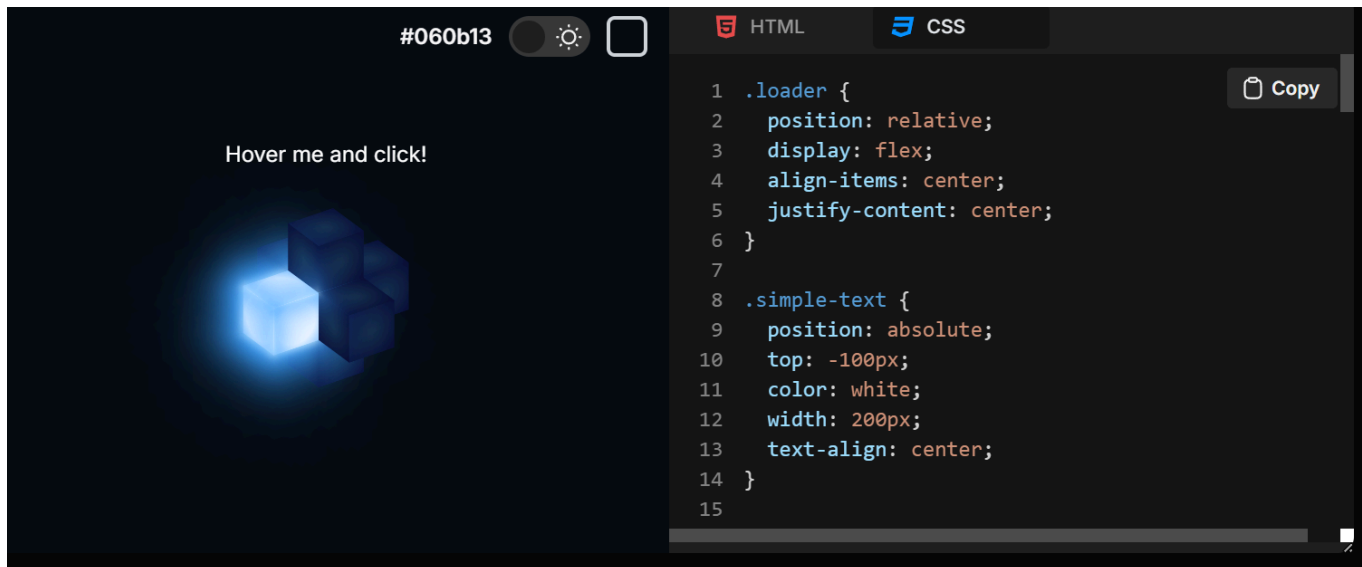
## 1. Loading experience — replace initial loader with Cube Loader

**Problem:** After the globe/card template lazy-loads, there's a bare 5–6s gap with nothing engaging on screen.

### Fix:

- Remove the current initial `StepLoader` entirely.
- Integrate the cube-loader template found at `template_repos/textura_templates/Cube_loader` (exact HTML/CSS pasted there — port to a React component, don't reference the raw files at runtime).
- Recolor to match the page background `#060b13` / `#0A0A0A` base + green/cyan accents per the design system — the reference screenshot's blue glow is close, keep that hue family.
- Behavior: cubes rotate continuously; **one cube glows at a time automatically** (no hover required — hover still works as an enhancement, doesn't gate the effect); **each cube "goes in" one at a time automatically** (no click required — click still works as an enhancement).
- Below the cube stack, show rotating short status text (e.g. "loading the globe...", "waking up the mind map...", "compiling agentic systems...") — purely cosmetic, ties to nothing real, just keeps the user's eyes busy.

- Duration: hold this screen 7–10s or until the globe + card template are actually ready, whichever is longer — never cut it short and reveal an unfinished scene.
- While this loader is showing, kick off the mind map background-preload (see Resolved Decisions).

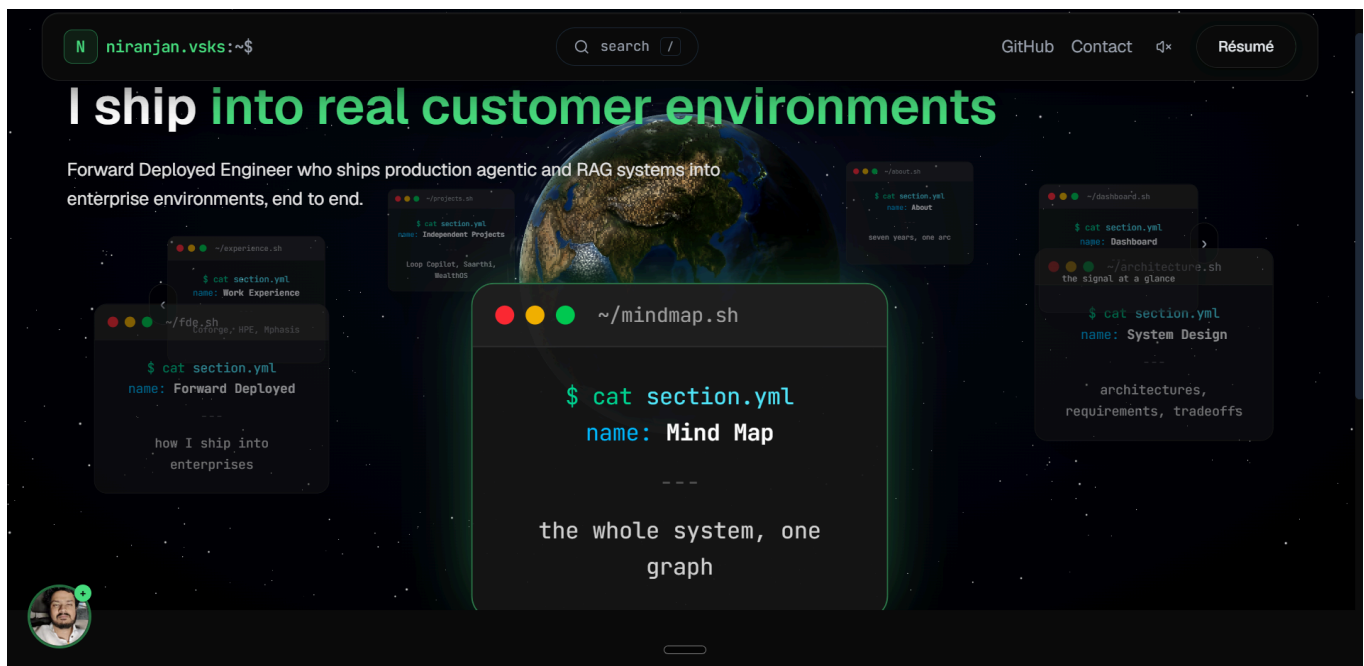


## 2. Landing page — center terminal card focus/blur behavior

**Problem:** Center card's "expand on focus" effect currently overshoots and its zoom needs slight reduction. Also want a glossy-corner glow when a card is centered/in-focus, fading out as a card drifts away from center.

### Fix:

- When a card rotates into the center focus position: expand it *slightly less* than current (reduce the scale delta — subtle, not dramatic).
- Add a glossy highlight effect on the card's corners/border when it's in focus (soft light sweep or corner glow, consistent with the dark terminal-card aesthetic — no white blowout).
- As a card rotates out of focus, the glow/gloss should fade out gradually (opacity transition tied to its distance from center), not cut off abruptly.

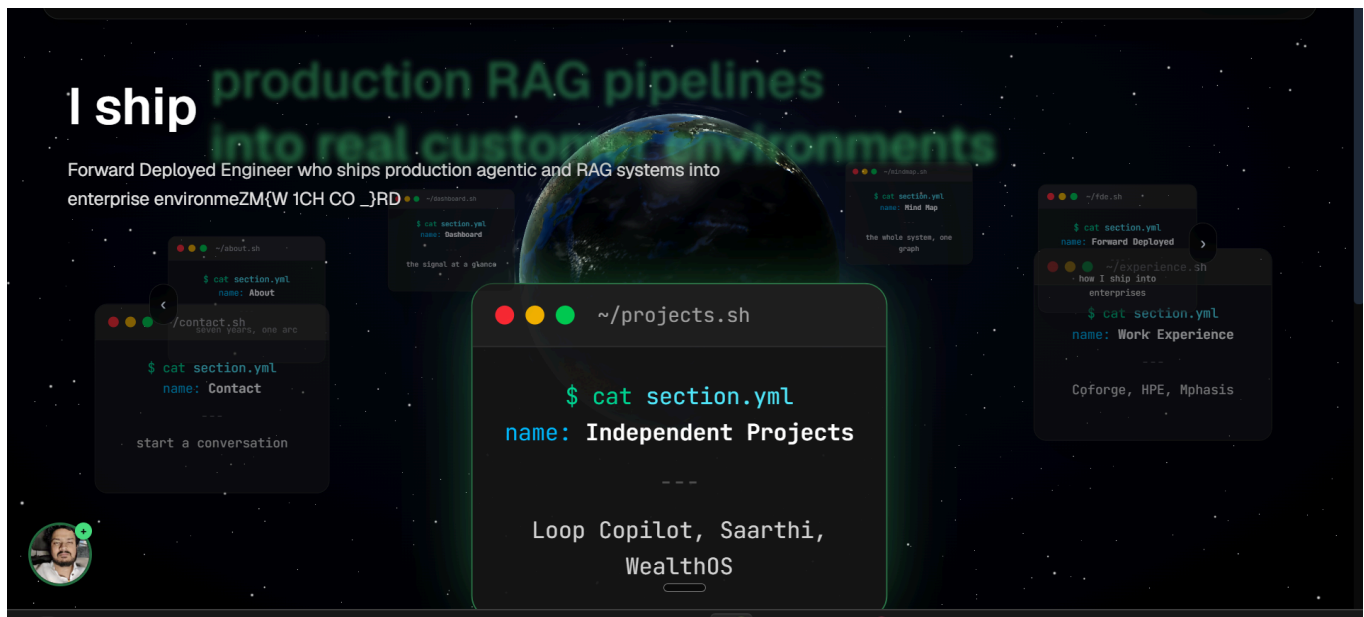


### 3. Landing page — header text overlap bug (critical, reported from live domain)

**Problem:** The "I ship [flip text]" header + summary line overlaps the globe/orbiting cards after the flip-word cycle runs; also glitches visibly during the transition. Looks broken/cheap on the deployed domain.

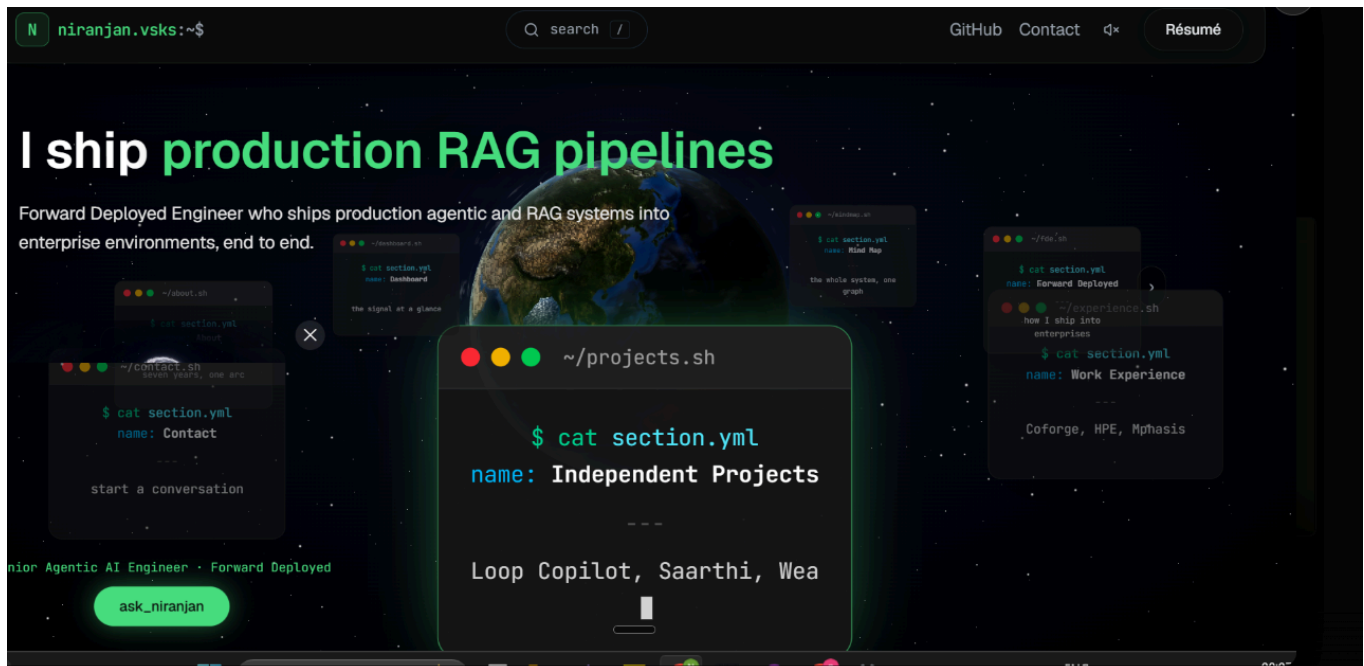
#### Fix:

- Header + summary text should **appear once on load**, then **auto-collapse after ~5s** (not the current behavior).
- Add a small arrow/chevron icon near the header position — clicking it re-expands the text.
- The header block must **never overlap** the globe or orbiting cards in either its expanded or collapsed state — reserve layout space or reposition so cards/globe are never occluded.
- Eliminate the glitch/flash during the flip-word transition and during collapse/expand — transitions must be smooth, no layout jump.



### 3.1 Landing page — avatar/dialogue glitch during loader, seamless background render, Starman preload

**Problem:** While cube loader (section 1) runs, avatar collapsible icon overlaps orbiting cards/terminal windows underneath — dialogue box bleeds through instead of staying hidden until loader finishes. Avatar itself doesn't render at all during the loading phase, then pops in glitchy.



#### Fix:

- While cube loader is active, avatar/collapsible icon and its dialogue box must stay fully hidden (zero opacity, zero pointer-events) — not mid-fade, not partially rendered, not overlapping cards behind it.

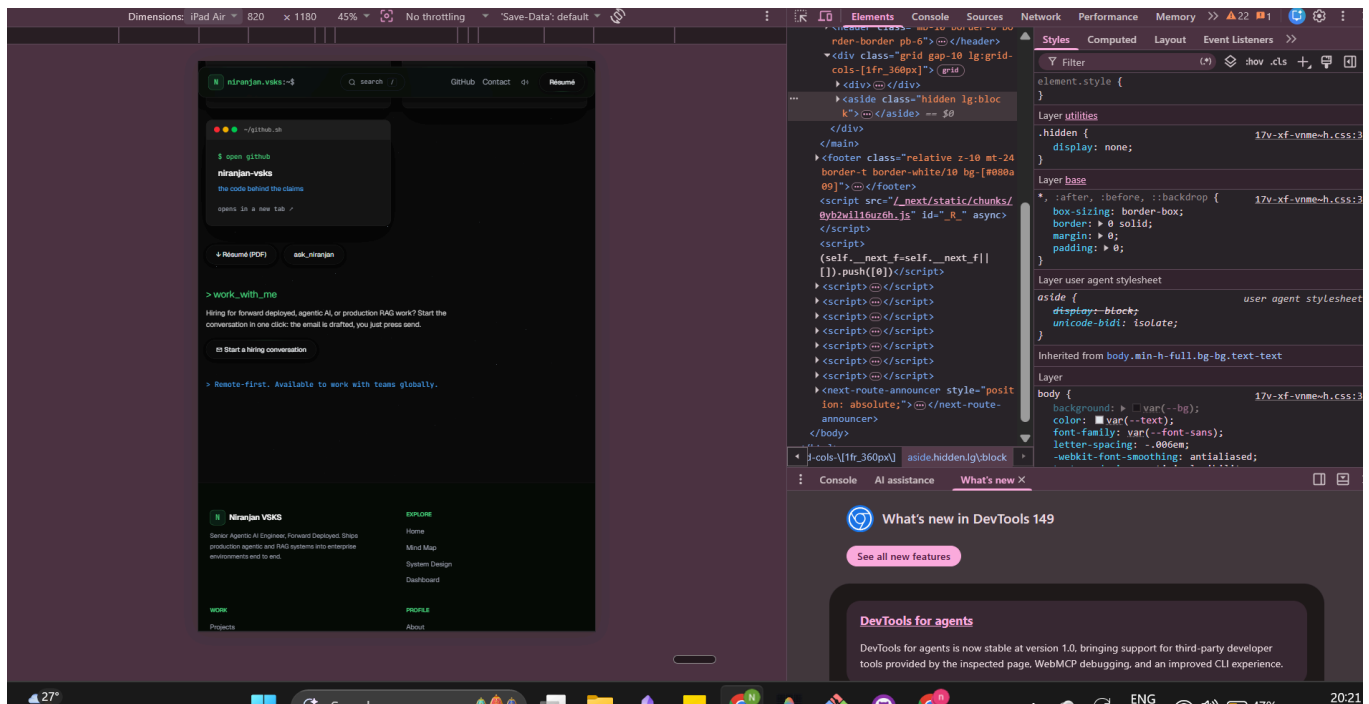
- All heavy templates (globe, orbiting terminal cards, avatar video, WebGL/animations) must finish mounting and positioning **behind** the cube loader, fully in background, before the loader dismisses — so the reveal is a single clean crossfade, not a cascade of popping/glitching elements.
- Avatar reveal must be synced to loader-complete, not independent — no separate late pop-in after cards already settled.
- Apply same background-preload treatment to the Contact page Starman video: preload/prime the video element in background (poster frame shown instantly, video buffering silently) so it's ready the instant Contact page paints, no missing/blank avatar.

## 4. Mobile — contact page avatar missing

**Problem:** On mobile, the Contact page's Starman avatar doesn't render at all.

**Fix:**

- Add a mobile fallback: static image (poster frame, not video) of Starman, always visible on mobile contact page. Doesn't need to animate.



## 5. Mobile — forced landscape / broken portrait cards

**Problem:** Visiting the desktop site on mobile currently forces landscape orientation to view properly; in portrait, cards don't load/layout correctly.

**Fix:**

- Portrait mode must be a first-class supported layout — cards render and stack correctly without requiring rotation.
- Audit and fix whatever viewport/orientation constraint is currently forcing landscape.
- Lazy-load transitions on mobile must stay smooth — optimize for latency (compression, defer non-critical JS) without dropping visual quality — no resolution or asset-quality degradation anywhere, not even by one pixel.

## 6. Chatbot — window controls (minimize / maximize / close)

**Problem:** Chat window currently opens as a partial/half-cut panel with no real window controls; the terminal-style traffic-light buttons are decorative only.

**Fix:**

- Wire up all three macOS-style traffic-light buttons on the chat window's terminal chrome to real behavior:
  - **Red** → Close the chat window.
  - **Yellow** → Minimize (collapse to a small pill/icon, re-expand on click).
  - **Green** → Maximize (expand to a larger, fully-visible chat view — not full-screen takeover, but noticeably larger and not clipped).
- On hover over each button, show a small tooltip label: "Close", "Minimize", "Maximize" — mimic native macOS window-control hover behavior, buttons can grow slightly on hover.
- Fix the current clipping bug where the chat window renders half off-screen — it must always render fully within viewport bounds at every window state.

## 7. Chatbot — remove leaked internal placeholder text from responses

**Problem:** Chatbot responses currently end with visible text referencing the Groq key / "will be integrated when actual key is given" — an internal placeholder leaking into user-facing output.

**Fix:**

- Remove this string entirely from anything the model or wrapper appends to responses. No internal build-state commentary should ever be user-visible.

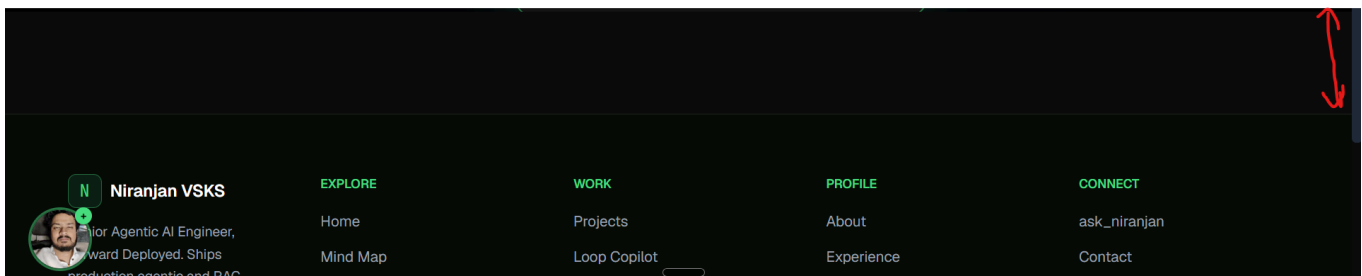
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## 8. Chatbot — guardrails via Markdown system prompts + model upgrade

## Fix (per resolved decisions above):

- Add a small set of versioned `.md` files acting as the system prompt / guardrail layer (persona, refusal rules, scope boundaries, tone). No RAG pipeline — this is intentionally lightweight.
- Route chat completions through **Vercel AI Gateway**, default model **Gemini 3**, fallback **Groq** — swap only the provider/model config, keep the existing call structure and surrounding code as-is.

Also in the footer Nav bar , this height marked in Arrow can be reduced a little so that we can accomodate more space for landing page components and they can be a little spaciouly shown (ever so slightly , not too much)

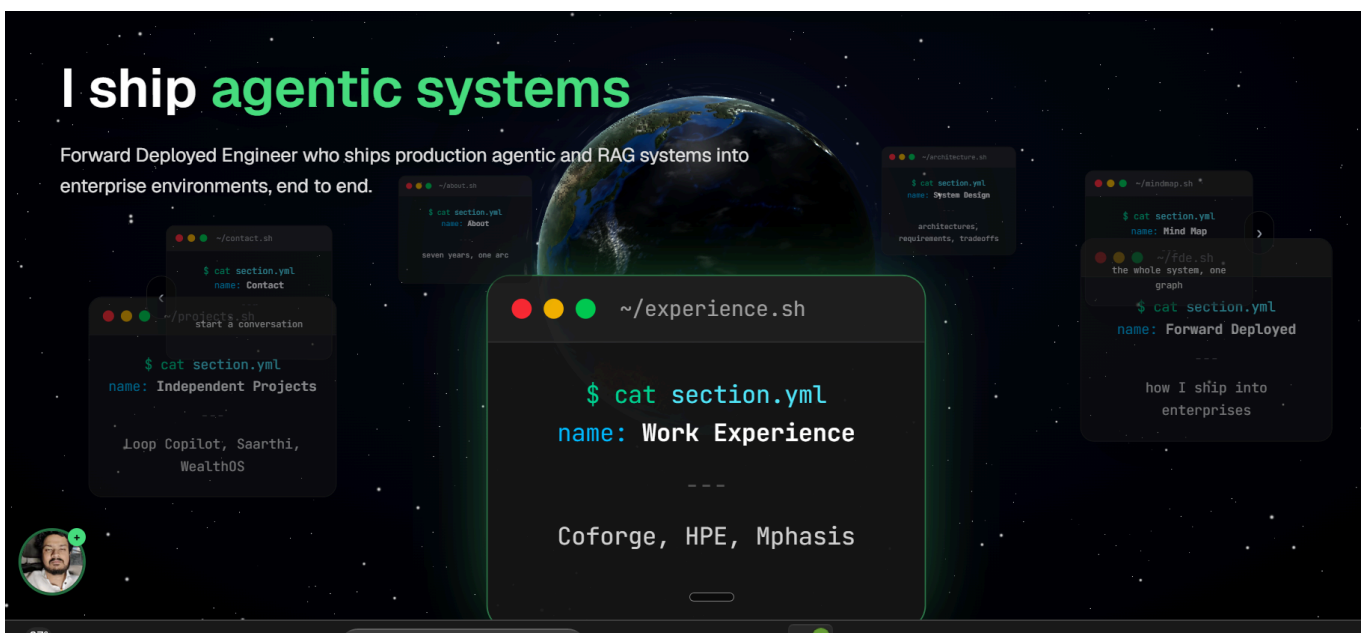


## 9. Landing page — sideways scroll for orbit cards

**Problem:** Cards orbit but there's no manual sideways scroll control beyond the existing arrows.

**Fix:**

- Enable horizontal scroll/swipe (trackpad, touch, and mouse-wheel-shift) to manually rotate/revolve the card carousel, on both desktop and mobile — in addition to the existing arrow controls, not replacing them.



## 10. Mind map — transition timing from keyword/skill links

**Problem:** Navigating to the mind map from a skill keyword currently snaps in ~1s, feels abrupt.

**Fix:**

- When arriving at the mind map via a keyword/skill deep-link, lengthen the entry transition to **2–2.5s** (currently ~1s). Direct navigation to `/map` (not via keyword) can keep its current timing.

## 11. Projects — HPE cross-tagging + Emphasis project pages

**Problem A:** The National Census chatbot project is tagged under HPE in its own project page, but the HPE experience section doesn't reference/link it back.

**Fix A:** Add a reference/link to the National Census chatbot project from the HPE section of `/experience` (or wherever HPE's work is listed), so the relationship is visible from both directions.

**Problem B:** "Mphasis" projects (older/less-relevant-to-current-role projects) currently have no dedicated page.

**Fix B:** Give Mphasis projects their own project page(s), consistent with the existing project page template/design system. Surgical, seamless — don't restructure the projects data model beyond adding these entries. System design and FDE Sections are not Required neither are wireframes and Slider Spectra , Just Heading of it with Flow Wave Green Background as the other project pages and the Description of it . with keywords thats it .

And Remove the "Remote-first, available to work with teams globally" in the Footer. Also in the Contact page , Add that but make it sound a little more professional.

added the sound effect of the typing typewriter effect to the public slash typing sound so public slash sounds folder so i want you to use that so in the project section in the independent project section wherever the typewriter effect is being shown or so wherever the typewriter effect is being shown I want you to play it, so note that even though multiple cards are showing the typewriter effect, we should not play it in unison, it should be played only once, like no chorus kind of sound effect should be there.